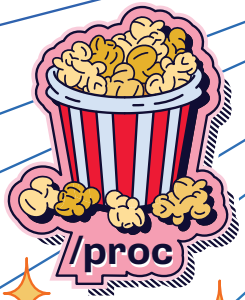
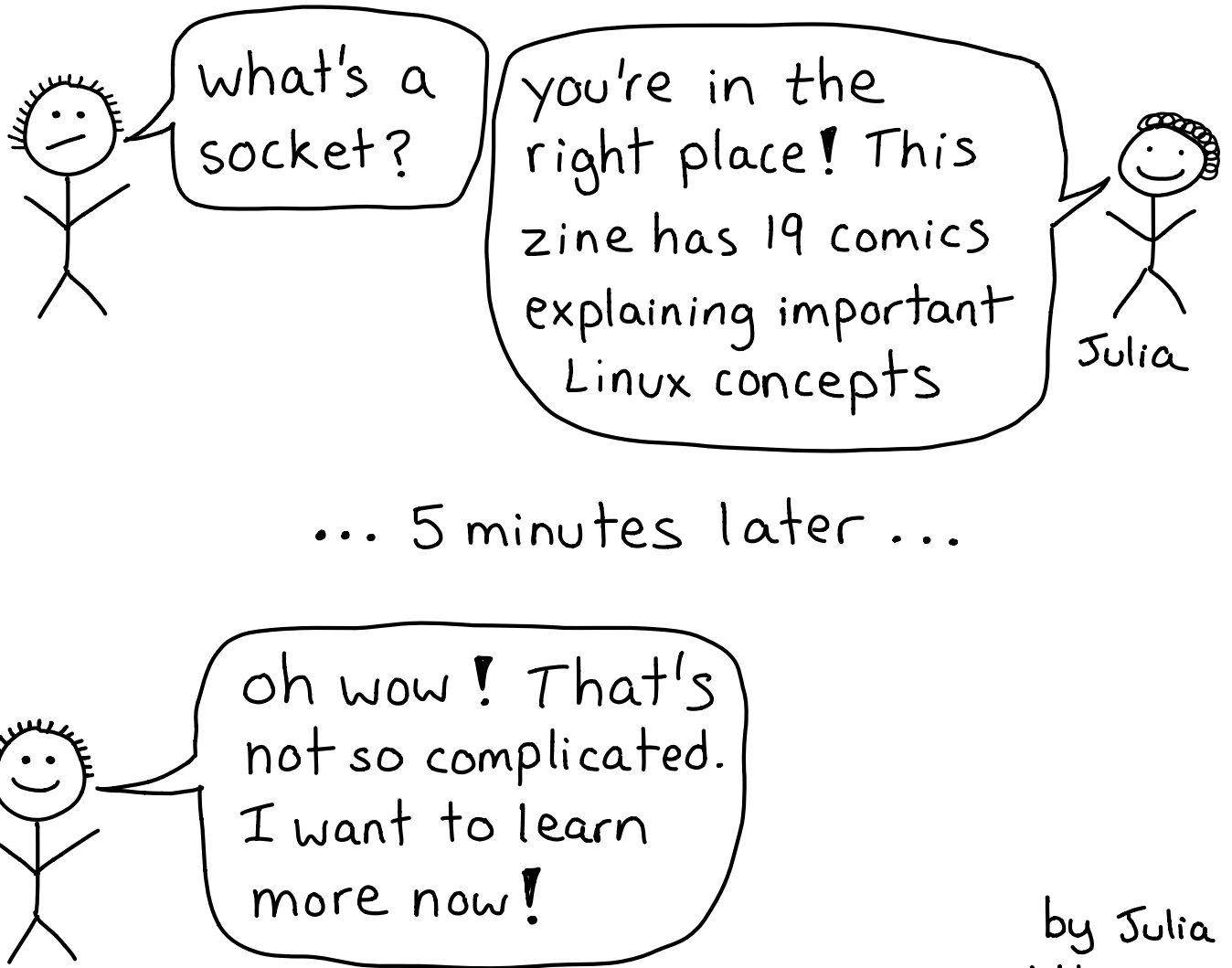
# Bite Size Linux

By Julia Evans




what's a socket?
you're in the right place! This zine has 19 comics explaining important Linux concepts
Julia
... 5 minutes later ...
oh wow! That's not so complicated. I want to learn more now!
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# unix permissions



There are 3 things you can do to a file

read write execute

ls -l file.txt shows you permissions
Here's how to interpret the output:

rw- rw- r-- bork staff

bork (user) can read & write

staff (group) can read & write

ANYONE can read

File permissions are 12 bits

| setuid | setgid | | user | group | all |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 110 | 110 | 100 |
| sticky | | | rwx | rwx | rwx |

For files:
- r = can read
- w = can write
- x = can execute

For directories it's approximately:
- r = can list files
- w = can create files
- x = can cd into & access files

110 in binary is 6

So rw- r-- r--
= 110 100 100
= 6 4 4

chmod 644 file.txt means change the permissions to:

rw- r-- r--

Simple!

setuid affects executables

$ls -l /bin/ping

rws r-x r-x root root

this means ping always runs as root

setgid does 3 different unrelated things for executables, directories, and regular files

# an amazing directory: /proc



Every process on Linux has a PID (process ID) like 42.

In /proc/42, there's a lot of VERY USEFUL information about process 42

## /proc/PID/cmdline

command line arguments the process was started with

## /proc/PID/exe

symlink to the process's binary. magic: works even if the binary has been deleted!

## /proc/PID/environ

all of the process's environment variables

## /proc/PID/status

Is the program running or asleep? How much memory is it using? And much more!

## /proc/PID/fd

Directory with every file the process has open!

Run `$ ls -l /proc/42/fd` to see the list of files for process 42.

These symlinks are also magic & you can use them to recover deleted files ♥

## /proc/PID/stack

The kernel's current stack for the process. Useful if it's stuck in a system call

## /proc/PID/maps

List of process's memory maps. Shared libraries, heap, anonymous maps, etc.

and more

Look at

**man proc**

for more information!

# system calls



The Linux kernel has code to do a lot of things

read from a hard drive

make network connections

create new process

kill process

change file permissions

keyboard drivers

your program doesn't know how to do those things

TCP? dude I have no idea how that works

NO I do not know how the ext4 filesystem is implemented I just want to read some files

programs ask Linux to do work for them using =system calls=

program: please write to this file

<switch to running kernel code>

Linux: done! I wrote 1097 bytes♥

<program resumes>

every program uses system calls

Python program: I use the 'open' syscall to open files

Java program: me too!

C program: me three!

and every system call has a number (eg chmod is #90 on x86_64)

So what's actually going on when you change a file's permissions is

program: run syscall #90 with these arguments

Linux: ok!

you can see which system calls a program is using with strace

```
$ strace ls /tmp
```

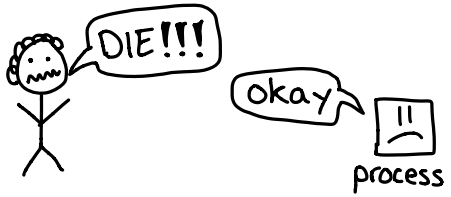
will show you every system call 'ls' uses! it's really fun!

strace is high overhead don't run it on your production database

# signals



If you've ever used kill you've used signals




the Linux kernel sends your process signals in lots of situations

- your child terminated
- that pipe is closed
- illegal instruction
- the timer you set expired
- Segmentation fault

you can send signals yourself with the kill system call or command

| | | |
|---|---|---|
| SIGINT | ctrl-C | various levels of "die" |
| SIGTERM | kill | |
| SIGKILL | kill -9 | |

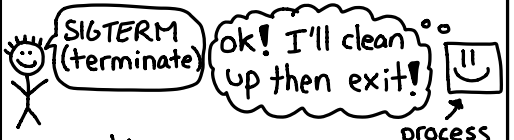
SIGHUP kill -HUP

often interpreted as "reload config", eg by nginx

Every signal has a default action, one of:

- ignore
- Kill process
- kill process AND make core dump file
- stop process
- resume process

Your program can set custom handlers for almost any signal

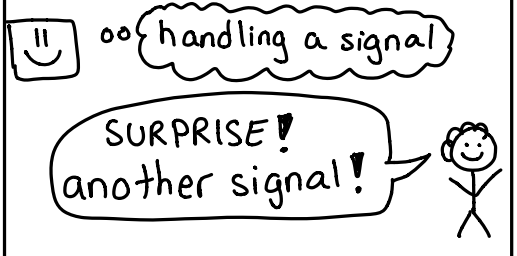



exceptions:

SIGSTOP & SIGKILL can't be ignored

got SIGKILLED

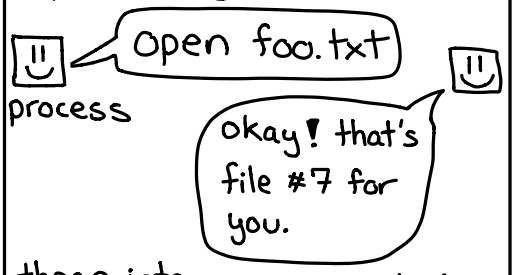
signals can be hard to handle correctly since they can happen at ANY time



# file descriptors



Unix systems use integers to track open files




these integers are called file descriptors

lsof (list open files) will show you a process's open files

```
$ lsof -p 4242   <- PID we're interested in
FD   NAME
 0   /dev/pts/tty1
 1   /dev/pts/tty1
 2   pipe:29174
 3   /home/bork/awesome.txt
 5   /tmp/
```

FD is for file descriptor

file descriptors can refer to:

- files on disk
- pipes
- sockets (network connections)
- terminals (like xterm)
- devices (your speaker! /dev/null!)
- LOTS MORE (eventfd, inotify, signalfd, epoll, etc etc)

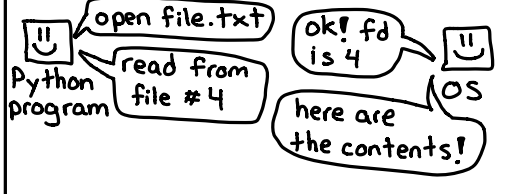
When you read or write to a file/pipe/network connection you do that using a file descriptor

Let's see how some simple Python code works under the hood:

Python:

```
f = open("file.txt")
f.readlines()
```

Behind the scenes:




(almost) every process has 3 standard FDs

stdin → 0
stdout → 1
stderr → 2

"read from stdin" means "read from the file descriptor 0"

could be a pipe or file or terminal

# pipes

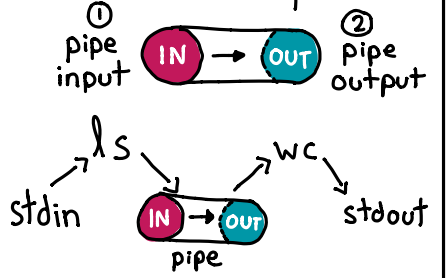


Sometimes you want to send the output of one process to the input of another

```
$ ls | wc -l
53
```

53 files!

a pipe is a pair of 2 magical file descriptors




When ls does write(IN, "hi")

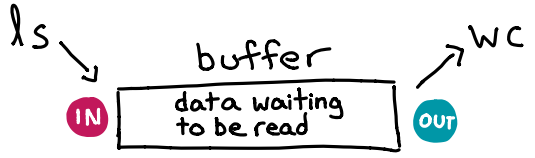
wc can read it!

read(OUT)

→ "hi"

Pipes are one-way. → You can't write to OUT.

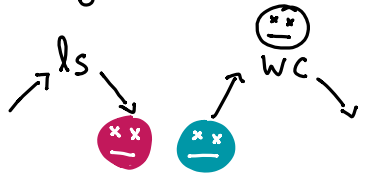
Linux creates a buffer for each pipe




If data gets written to the pipe faster than it's read, the buffer will fill up. IN OUT

When the buffer is full, writes to IN will block (wait) until the reader reads. This is normal & ok

what if your target process dies?




If wc dies, the pipe will close and ls will be sent SIGPIPE. By default SIGPIPE terminates your process.

## named pipes

```
$ mkfifo my-pipe
```

This lets 2 unrelated processes communicate through a pipe!

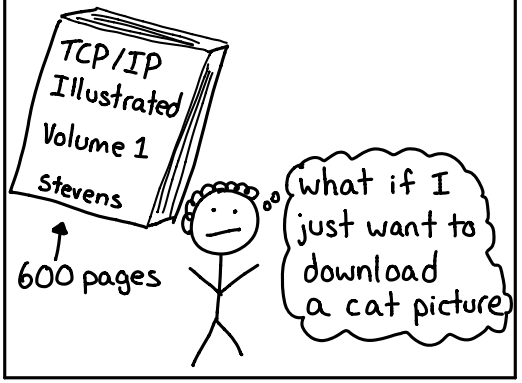
```
f=open(./my-pipe)
f.write("hi!\n")
```

```
f=open(./my-pipe)
f.readline() ← "hi!"
```

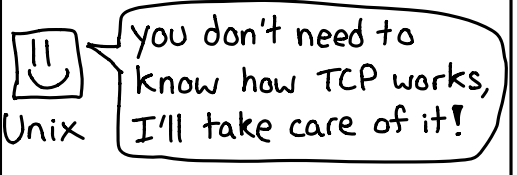
# sockets



networking protocols are complicated




Unix systems have an API called the "socket API" that makes it easier to make network connections (Windows too! :))




here's what getting a cat picture with the socket API looks like:

① Create a socket

```
fd = socket(AF_INET, SOCK_STREAM ...
```

② Connect to an IP/port

```
connect(fd, 12.13.14.15:80)
```

③ Make a request

```
write(fd, "GET /cat.png HTTP/1.1 ...
```

④ Read the response

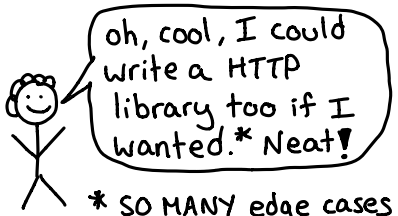
```
cat_picture = read(fd ...
```

Every HTTP library uses sockets under the hood

```
$ curl awesome.com
```

sockets

Python: `requests.get("yay.us")`




## AF_INET? What's that?

AF_INET means basically "internet socket": it lets you connect to other computers on the internet using their IP address.

The main alternative is AF_UNIX ("unix domain socket") for connecting to programs on the same computer

## 3 kinds of internet (AF_INET) sockets:

SOCK_STREAM = TCP
↑ curl uses this

SOCK_DGRAM = UDP
↑ dig (DNS) uses this

SOCK_RAW = just let me send IP packets I will implement my own protocol
↑ ping uses this

# unix domain sockets



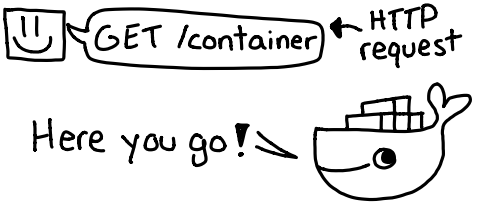
unix domain sockets are files.

```
$ file mysock.sock
socket
```

the file's permissions determine who can send data to the socket

they let 2 programs on the same computer communicate.

Docker uses Unix domain sockets, for example!




There are 2 kinds of unix domain sockets:

stream: like TCP! Lets you send a continuous stream of bytes

datagram: like UDP! Send discrete chunks of data

## advantage 1

Lets you use file permissions to restrict access to HTTP/database services!

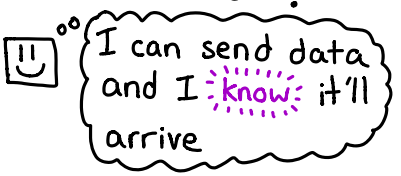
```
chmod 600 secret.sock
```

This is why Docker uses a unix domain socket

## advantage 2

UDP sockets aren't always reliable (even on the same computer).

unix domain datagram sockets are reliable! And won't reorder!




## advantage 3

You can send a file descriptor over a unix domain socket.

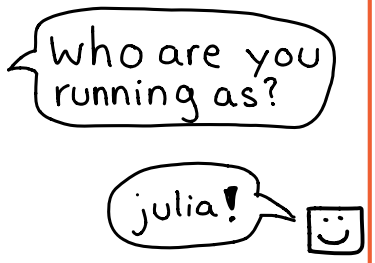
Useful when handling untrusted input files!

# what's in a process?



## PID

## USER and GROUP




## ENVIRONMENT VARIABLES

like PATH! you can set them with:

```
$ env A=val ./program
```

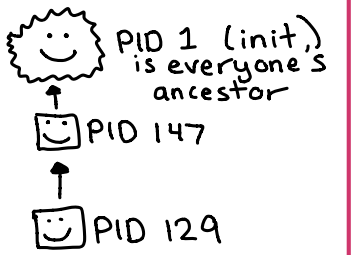
## SIGNAL HANDLERS

## WORKING DIRECTORY

Relative paths (./blah) are relative to the working directory!

chdir changes it.

## PARENT PID




## COMMAND LINE ARGUMENTS

See them in

/proc/PID/cmdline

## OPEN FILES

Every open file has an offset.

## MEMORY

heap! stack!

shared libraries!

the program's binary!

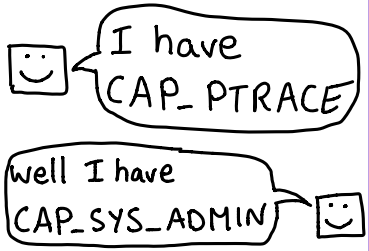
mmaped files!

## THREADS

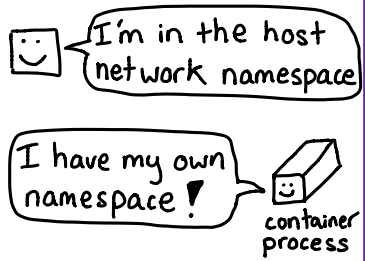
sometimes one

sometimes LOTS

## CAPABILITIES




## NAMESPACES



# threads



Threads let a process do many different things at the same time

process:

thread 1: I'm calculating ten million digits of π! so fun!

thread 2: I'm finding a REALLY BIG prime number!

threads in the same process share memory

thread 1: I'll write some digits of π to 0x129420 in memory

thread 2: uh oh that's where I was putting my prime numbers

and they share code

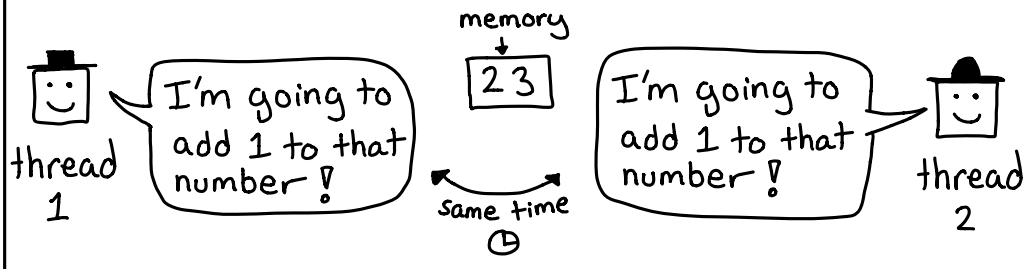
```
calculate_pi
find_big_prime_number
```

but each thread has its own stack and they can be run by different CPUs at the same time

CPU 1 — π thread

CPU 2 — primes thread

sharing memory can cause problems (race conditions!)




RESULT: 24 ← WRONG. Should be 25!

Why use threads instead of starting a new process?

→ a thread takes less time to create

→ sharing data between threads is very easy. But it's also easier to make mistakes with threads

thread 1: you weren't supposed to CHANGE that data!

# floating point



## a double is 64 bits

sign exponent fraction

1001101 1001101 1001101 1001101
1001101 1001101 1001101 1001101

$\pm 2^{E-1023} \times 1.\text{frac}$

That means there are $2^{64}$ doubles

The biggest one is about $2^{1023}$

## weird double arithmetic

$2^{52} + 0.2 = 2^{52}$ ← (the next number after $2^{52}$ is $2^{52}+1$)

$1 + \frac{1}{2^{54}} = 1$ ← (the next number after 1 is $1 + \frac{1}{2^{52}}$)

$2^{2000} = \text{infinity}$ ← infinity is a double

infinity - infinity = nan ← nan = "not a number"

doubles get farther apart as they get bigger

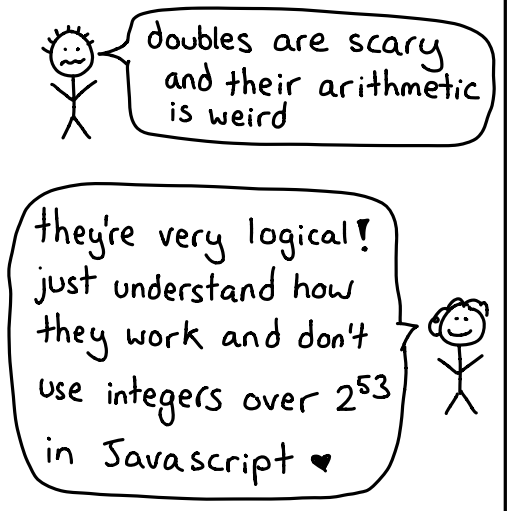
between $2^n$ and $2^{n+1}$ there are always $2^{52}$ doubles, evenly spaced

that means the next double after $2^{60}$ is $2^{60}+64$ ↖ $\frac{2^{60}}{2^{52}}$

Javascript only has doubles (no integers!)

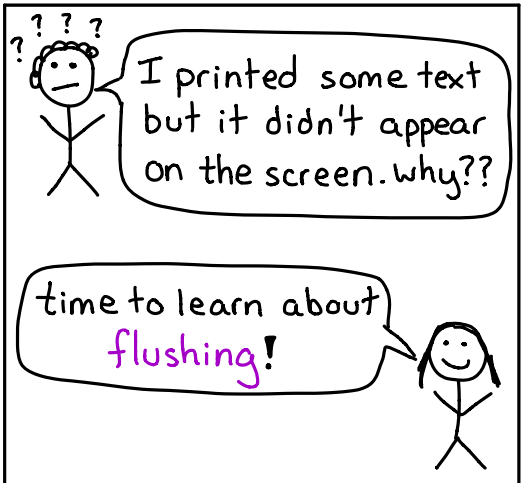
```
> 2**53
9007199254740992

> 2**53+1
9007199254740992
```

same number! uh oh!



# file buffering



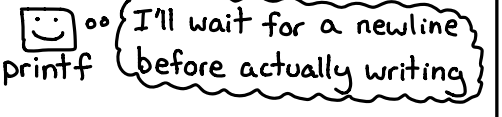



On Linux you write to files & terminals with a system call called

♥ write ♥

I/O libraries don't always call write when you print

```
printf("I ♥ cats");
```



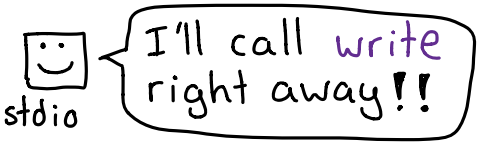

This is called buffering and it helps save on syscalls

3 kinds of buffering
(defaults vary by library)

1. None. This is the default for stderr
2. Line buffering. (write after newline). The default for terminals.
3. "full" buffering. (write in big chunks). The default for files and pipes.

## flushing

To force your IO library to write everything it has in its buffer right now, call flush!



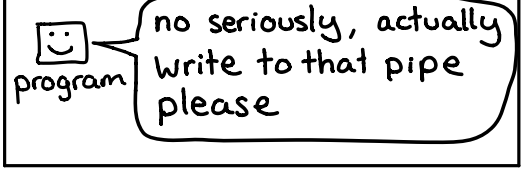

when it's useful to flush

→ when writing an interactive prompt!

Python example:

```
print("password: ", flush=True)
```

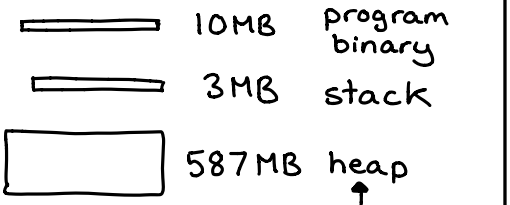
→ when you're writing to a pipe / socket



# memory allocation



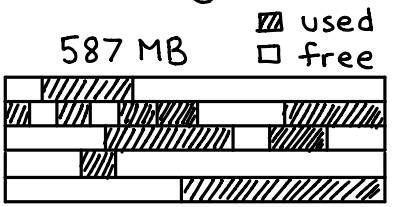
your program has memory



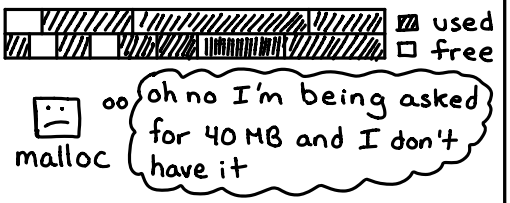

the heap is what your allocator manages

your memory allocator (malloc) is responsible for 2 things.

THING 1: keep track of what memory is used/free




THING 2: Ask the OS for more memory!




your memory allocator's interface

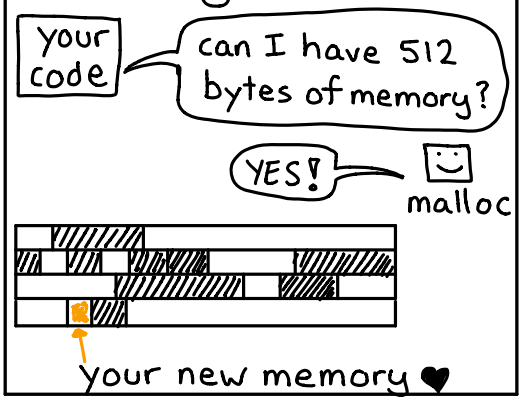
malloc (size_t size)
allocate size bytes of memory & return a pointer to it

free (void* pointer)
mark the memory as unused (and maybe give back to the OS)

realloc(void* pointer, size_t size)
ask for more/less memory for pointer

calloc (size_t members, size_t size)
allocate array + initialize to 0

malloc tries to fill in unused space when you ask for memory




malloc isn't magic

it's just a function!

you can always:

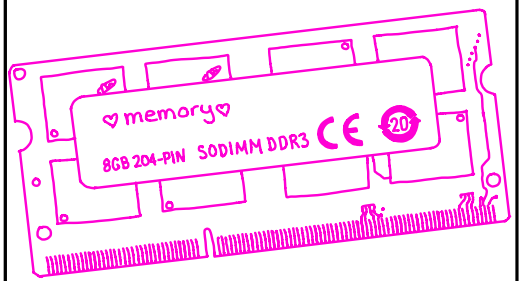
→ use a different malloc library like jemalloc or tcmalloc (easy!)

→ implement your own malloc (harder)

# virtual memory



your computer has physical memory



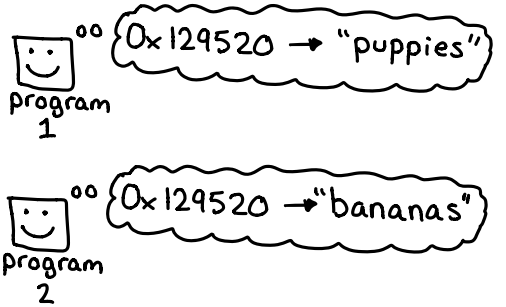

physical memory has addresses

0 - 8GB

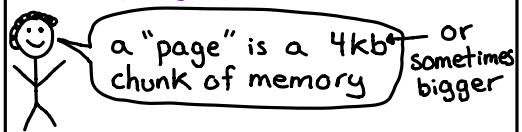
but when your program references an address like 0x5c69a2a2

that's not a physical memory address! It's a virtual address

every program has its own virtual address space

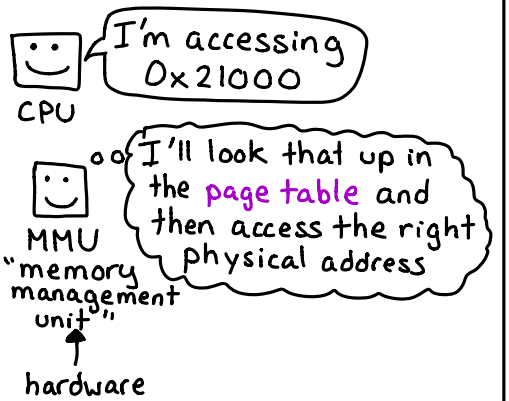



Linux keeps a mapping from virtual memory pages to physical memory pages called the "page table"




| PID | virtual addr | physical addr |
|---|---|---|
| 1971 | 0x20000 | 0x192000 |
| 2310 | 0x20000 | 0x228000 |
| 2310 | 0x21000 | 0x9788000 |

when your program accesses a virtual address




every time you switch which process is running, Linux needs to switch the page table

# shared libraries



Most programs on Linux use a bunch of C libraries

some popular libraries:

- openssl (for SSL!)
- sqlite (embedded db!)
- lib pcre (regular expressions!)
- zlib (gzip!)
- libstdc++ (C++ standard library!)

There are 2 ways to use any library

1. Link it into your binary

big binary with lots of things!

2. Use separate shared libraries

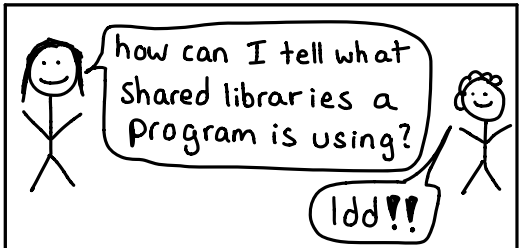
← all different files

Programs like this:

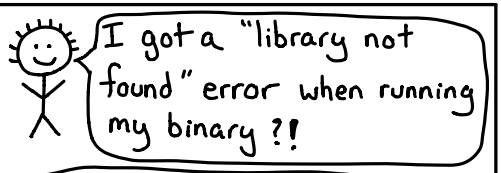
are called "statically linked"

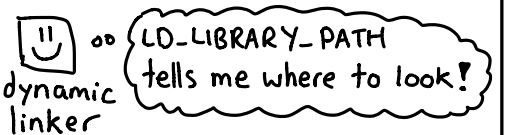
and programs like this:

are called "dynamically linked"




```
$ ldd /usr/bin/curl
        libz.so.1 => /lib/x86-64..
        libresolv.so.2 => ...
        libc.so.6 => ...
        +34 more
```




## Where the dynamic linker looks

1. DT_RPATH in your executable
2. LD_LIBRARY_PATH
3. DT_RUNPATH in executable
4. /etc/ld.so.cache (run ldconfig -p to see contents)
5. /lib, /usr/lib

# copy on write



On Linux, you start new processes using the fork() or clone() system call

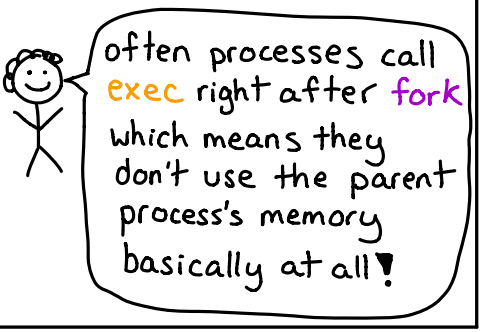
calling fork gives you a child process that's a copy of you

the cloned process has EXACTLY the same memory

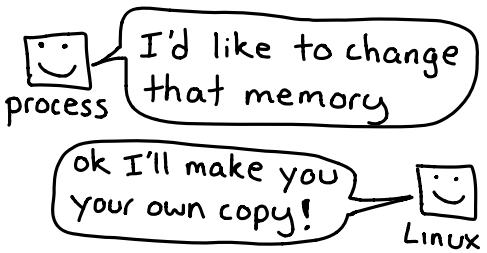
- → same heap
- → same stack
- → same memory maps

if the parent has 3GB of memory, the child will too

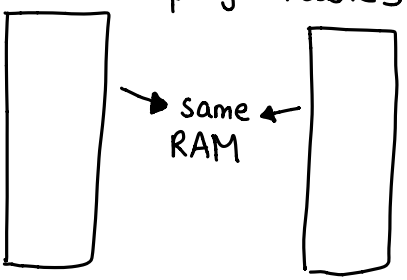
copying all that memory every time we fork would be slow and a waste of RAM




so Linux lets them share physical RAM and only copies the memory when one of them tries to write.



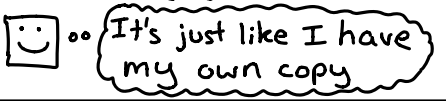

Linux does this by giving both the processes identical page tables




but marks every page as read only

when a process tries to write to a shared memory address

1. there's a page fault
2. Linux makes a copy of the page & updates the page table
3. the process continues, blissfully ignorant



# page faults



every Linux process has a page table

★ page table ★

| virtual memory address | physical memory address |
|---|---|
| 0x19723000 | 0x1422000 |
| 0x19724000 | 0x1423000 |
| 0x1524000 | not in memory |
| 0x1844000 | 0x4a000 read only |

some pages are marked as either

★ read only

★ not resident in memory

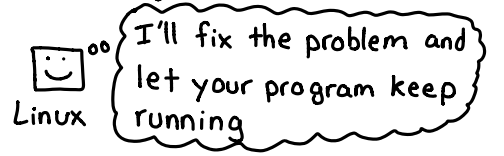
when you try to access a page that's marked "not in memory", that triggers a ! page fault !

What happens during a page fault?

→ the MMU sends an interrupt

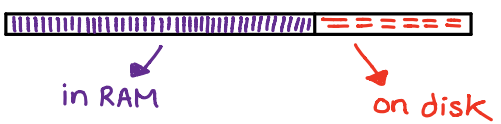
→ your program stops running

→ Linux kernel code to handle the page fault runs




"not in memory" usually means the data is on disk!

virtual memory




Having some virtual memory that is actually on disk is how swap and mmap work

## how swap works

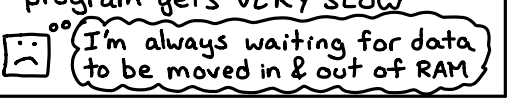
① run out of RAM

② Linux saves some RAM data to disk

③ mark those pages as "not resident in memory" in the page table

④ When a program tries to access the memory there's a ! page fault !

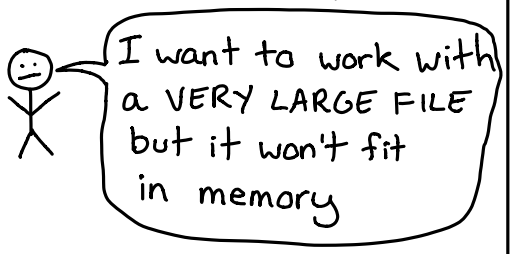
⑥ if this happens a lot your program gets VERY SLOW



# mmap



## what's mmap for?




## load files lazily with mmap

When you mmap a file, it gets mapped into your program's memory

but nothing is ACTUALLY read into RAM until you try to access the memory (how it works: page faults!)

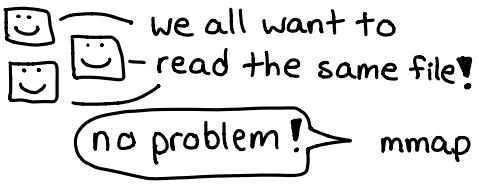
## how to mmap in Python

```
import mmap
f = open("HUGE.txt")
mm = mmap.mmap(f.fileno(), 0)
print(mm[-1000:])
```

this won't read the file from disk! Finishes ~instantly.

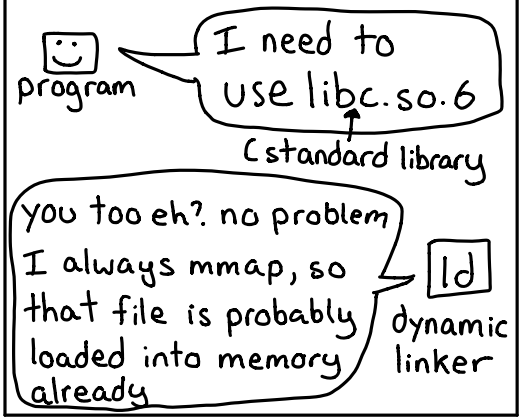
this will read only the last 1000 bytes!

## sharing big files with mmap




Even if 10 processes mmap a file, it will only be read into memory ♥once♥

## dynamic linking uses mmap




## anonymous memory maps

- not from a file (memory set to 0 by default)
- with MAP_SHARED, you can use them to share memory with a subprocess!

# man page sections



man pages are split up into 8 sections

① ② ③ ④ ⑤ ⑥ ⑦ ⑧

$ man 2 read

means "get me the man page for read from section 2"

There's both
- a program called "read"
- and a system call called "read"

so

$ man 1 read

gives you a different man page from

$ man 2 read

If you don't specify a section, man will look through all the sections & show the first one it finds

## man page sections

① programs

$ man grep
$ man ls

② system calls

$ man sendfile
$ man ptrace

③ C functions

$ man printf
$ man fopen

④ devices

$ man null
for /dev/null docs

⑤ file formats

$ man sudoers
for /etc/sudoers
$ man proc
files in /proc!

⑥ games

not super useful.
$ man sl
is funny if you have sl though.

⑦ miscellaneous

explains concepts!

$ man 7 pipe

$ man 7 symlink

⑧ sysadmin programs

$ man apt

$ man chroot

Want to learn more? I highly recommend this book →

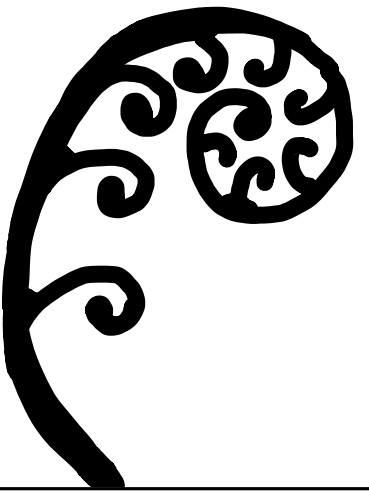
Every chapter is a readable, short (usually 10-20 page) explananation of a Linux system.

I used it as a reference constantly when writing this zine.

I ♡ it because even though it's huge and comprehensive (1500 pages!), the chapters are short and self-contained and it's very easy to pick it up and learn something.

# THE LINUX PROGRAMMING INTERFACE

MICHAEL KERRISK



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